Netra

Emergency Dispatched System

HORAMA AI Module



Problem Statement:

- Static signals do not adapt to real-time vehicle density, causing unnecessary congestion.
  Accidents are often missed or detected too late, leading to delayed emergency response.

GEOGRAPHIES IN DEPTH

India loses billions of rupees to traffic jams. Is ride-sharing the solution?

Oct 6, 2016

- India faces severe **traffic congestion**, leading to significant economic losses, fuel wastage, and increased pollution. Traditional traffic systems rely on fixed signal timings and lack real-time adaptability to changing traffic conditions.
- There is a need for a **smart, adaptive traffic management system** that dynamically optimizes signal timing based on real-time traffic flow to reduce congestion and improve overall efficiency.

Every second of delay in emergency response can cost a life. -Lack of priority systems prevents ambulances and fire trucks from moving swiftly through traffic.



Netra – Smart Emergency Response System

💰 **Economic loss: Traffic congestion in India costs approximately ₹3–5 lakh crore per year due to fuel waste, time loss, and reduced productivity.**

Proposed Solution

India records **over 1.7 lakh road deaths every year**, while traffic congestion costs billions in economic losses and fuel wastage. A major issue is the **lack of real-time coordination between traffic systems and emergency response**.

Netra solves this by using AI to dynamically adjust traffic signals based on vehicle density, instantly detect accidents, and automatically trigger emergency response. It identifies the nearest ambulance, optimizes routes, and creates a **green corridor** through smart signal control.

🔥 **Beyond Traffic Optimization → Full Emergency Response System**

🌍 **Environmental loss: Congestion significantly increases air pollution (CO_2, PM2.5) and fuel consumption, worsening urban air quality and public health.**

₹1.47 lakh crore per year (~$18–20 billion)

TOI

KEY FEATURES :-

🚨 **Real-Time Accident Detection + Instant Action**

🚑 **Automated Ambulance Dispatch with Smart Routing**

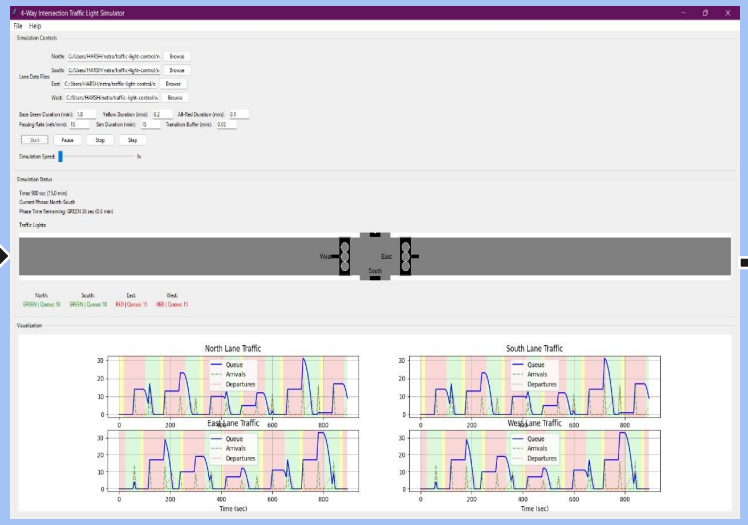
🟢 **Dynamic Green Corridor Creation**

👑 **VIP & Emergency Priority-Based Signal Control**

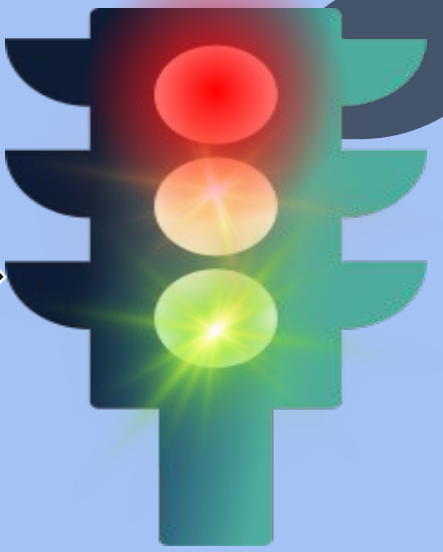
Model Specification

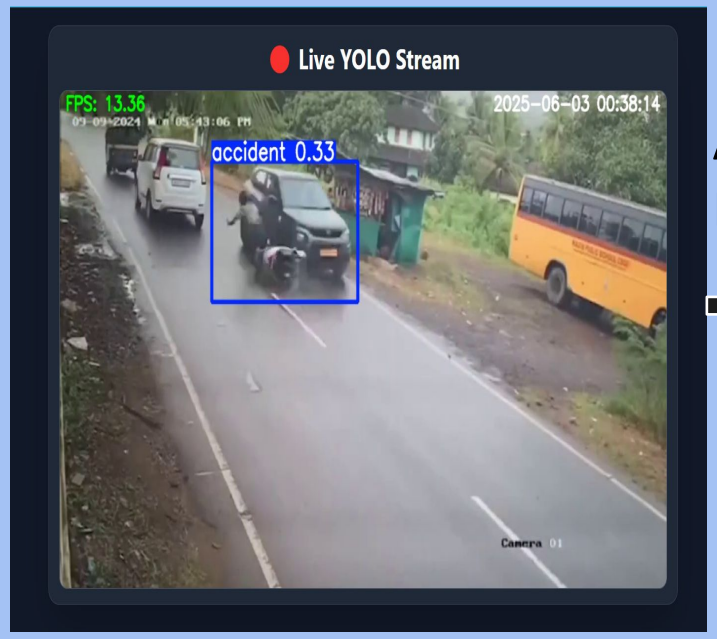


Analyzing data

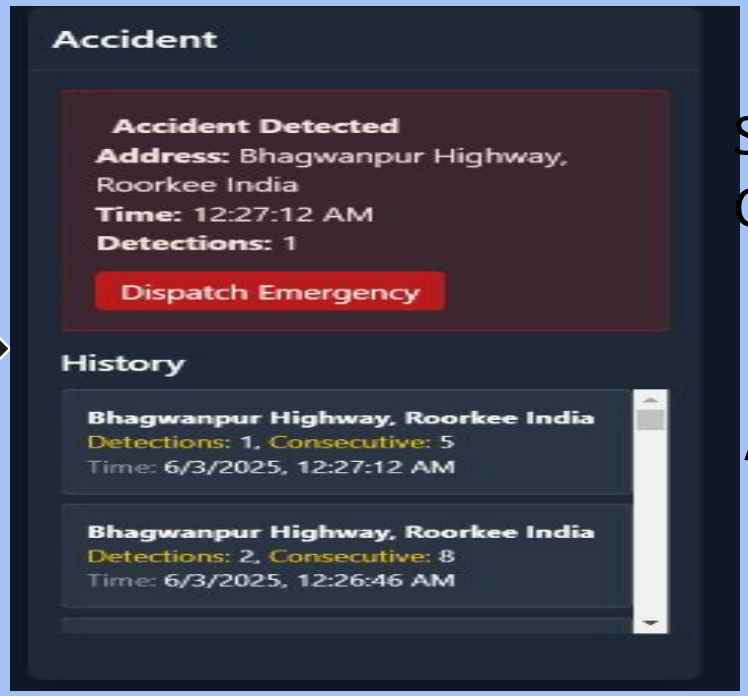


Adjust Traffic lights





Accident Detected



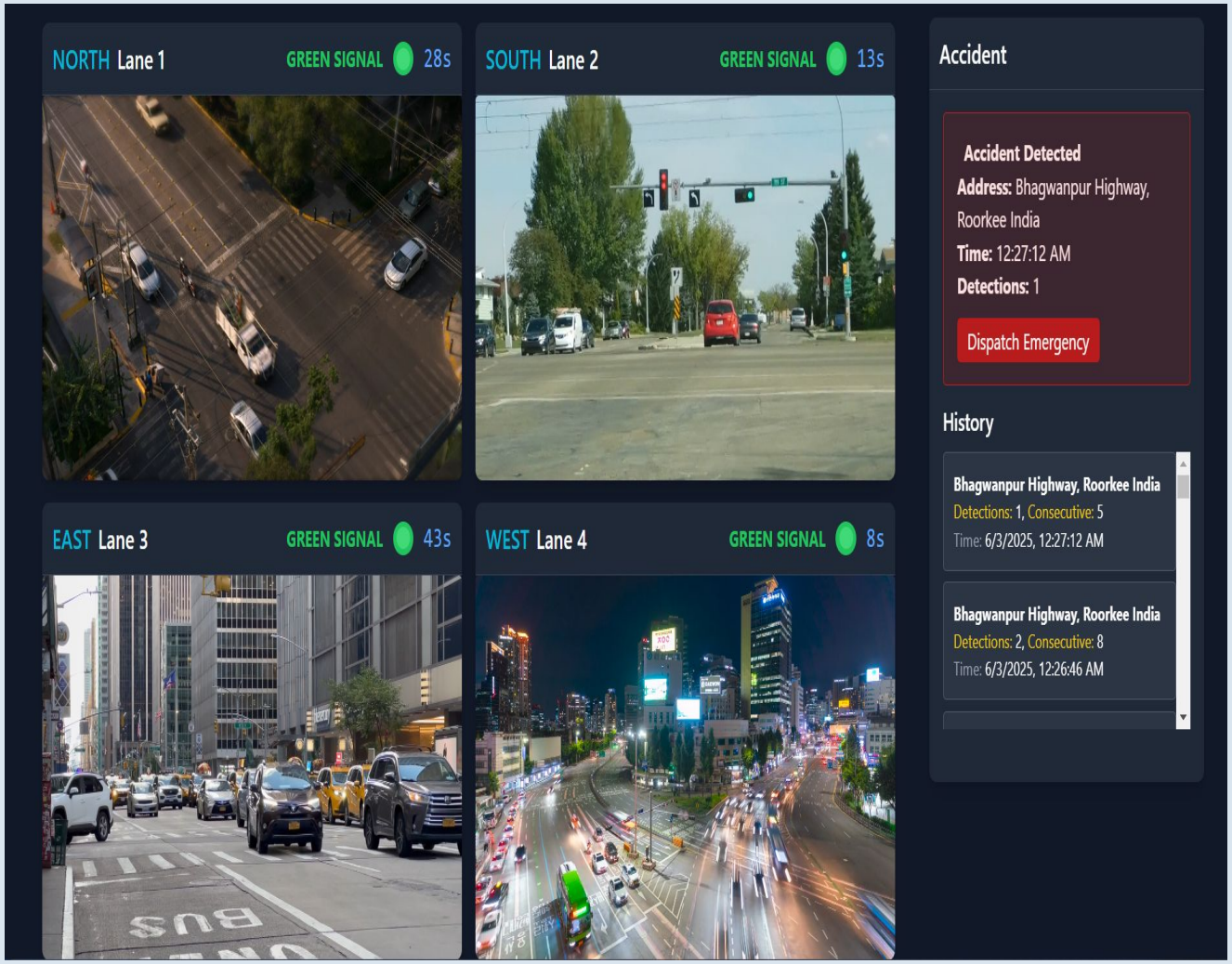
Status Confirmed

Ambulance Dispatched



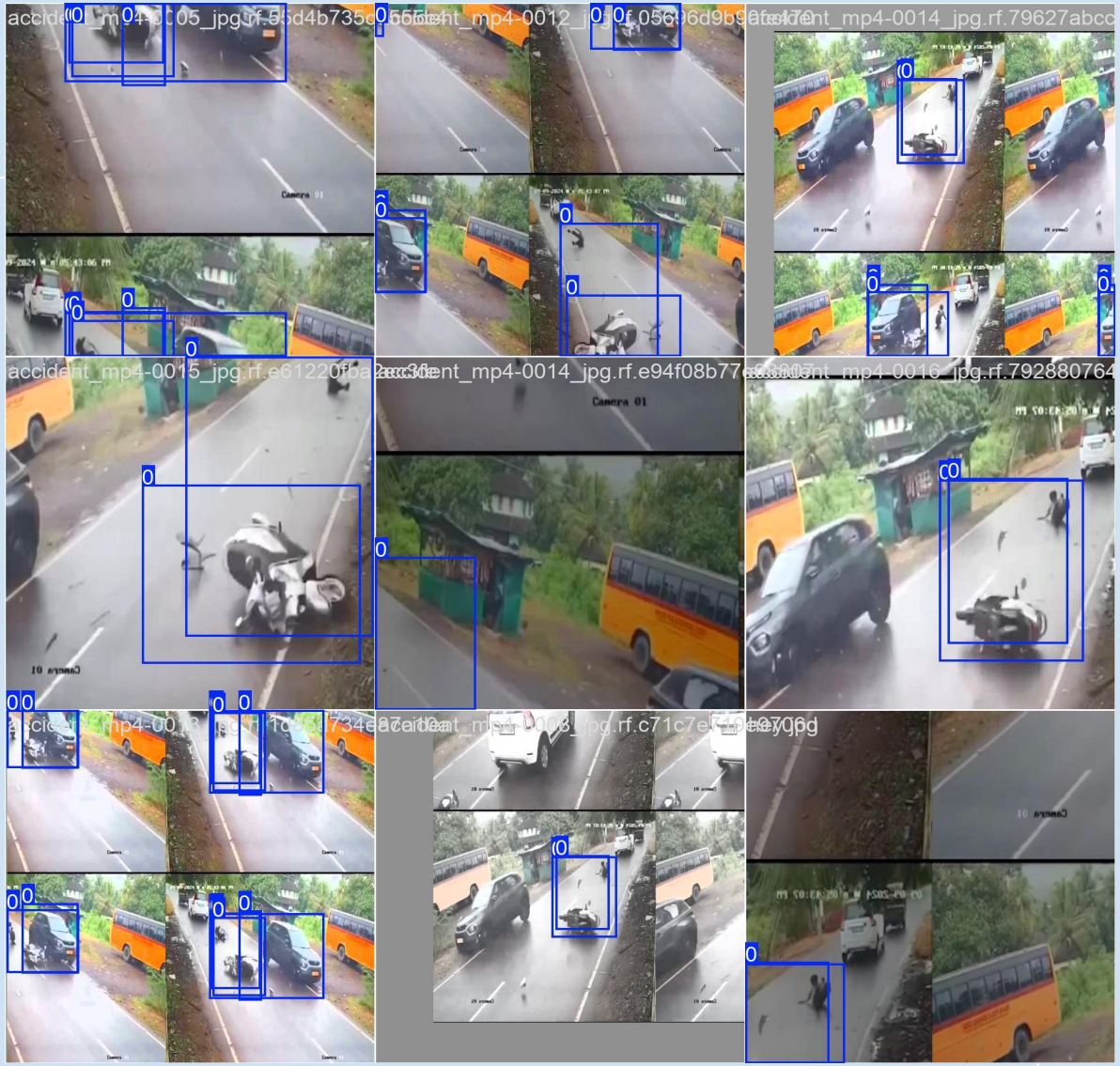
Prototype :

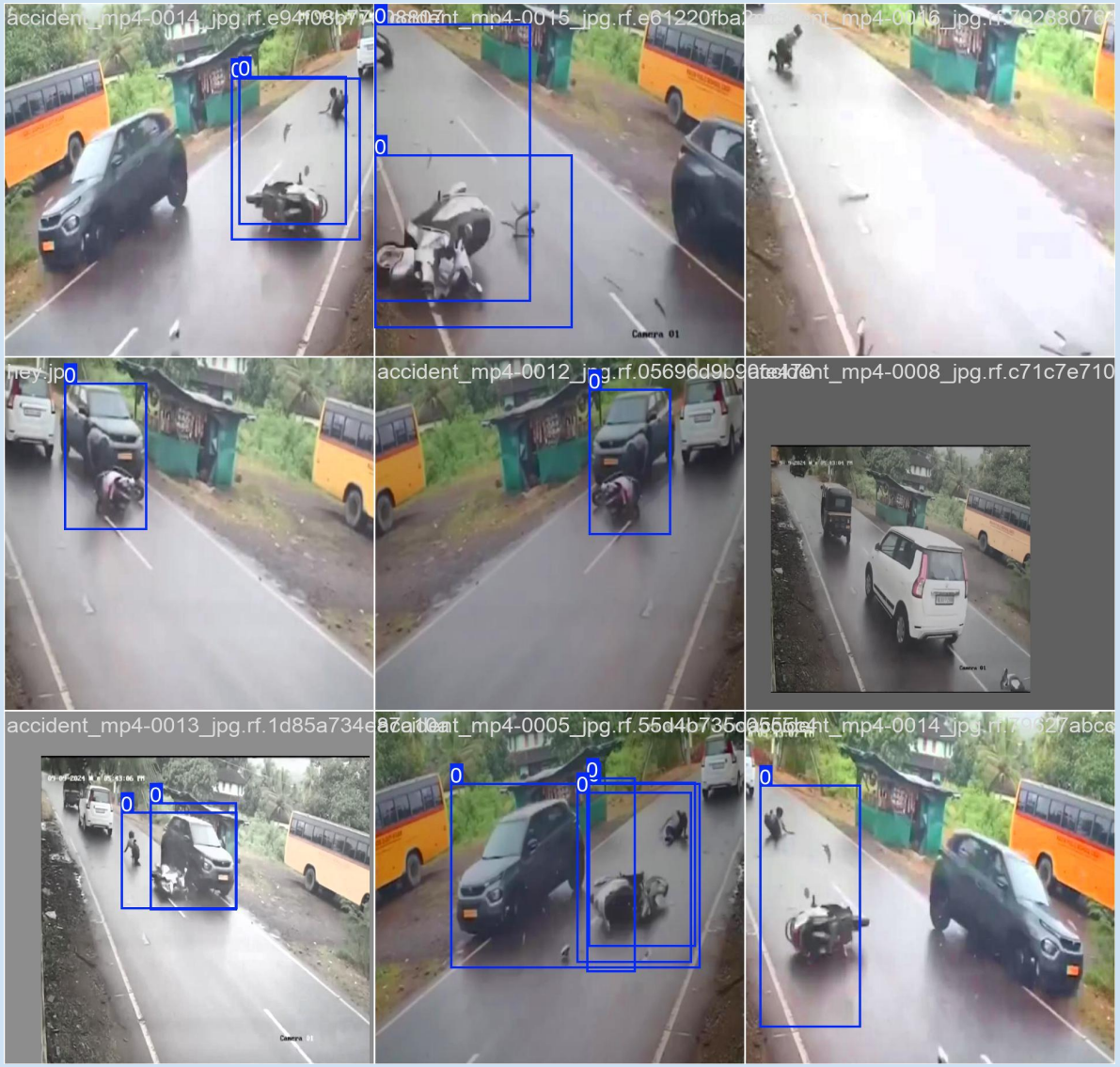
Tested on Virtual Environment





Trained Model Using real accidents



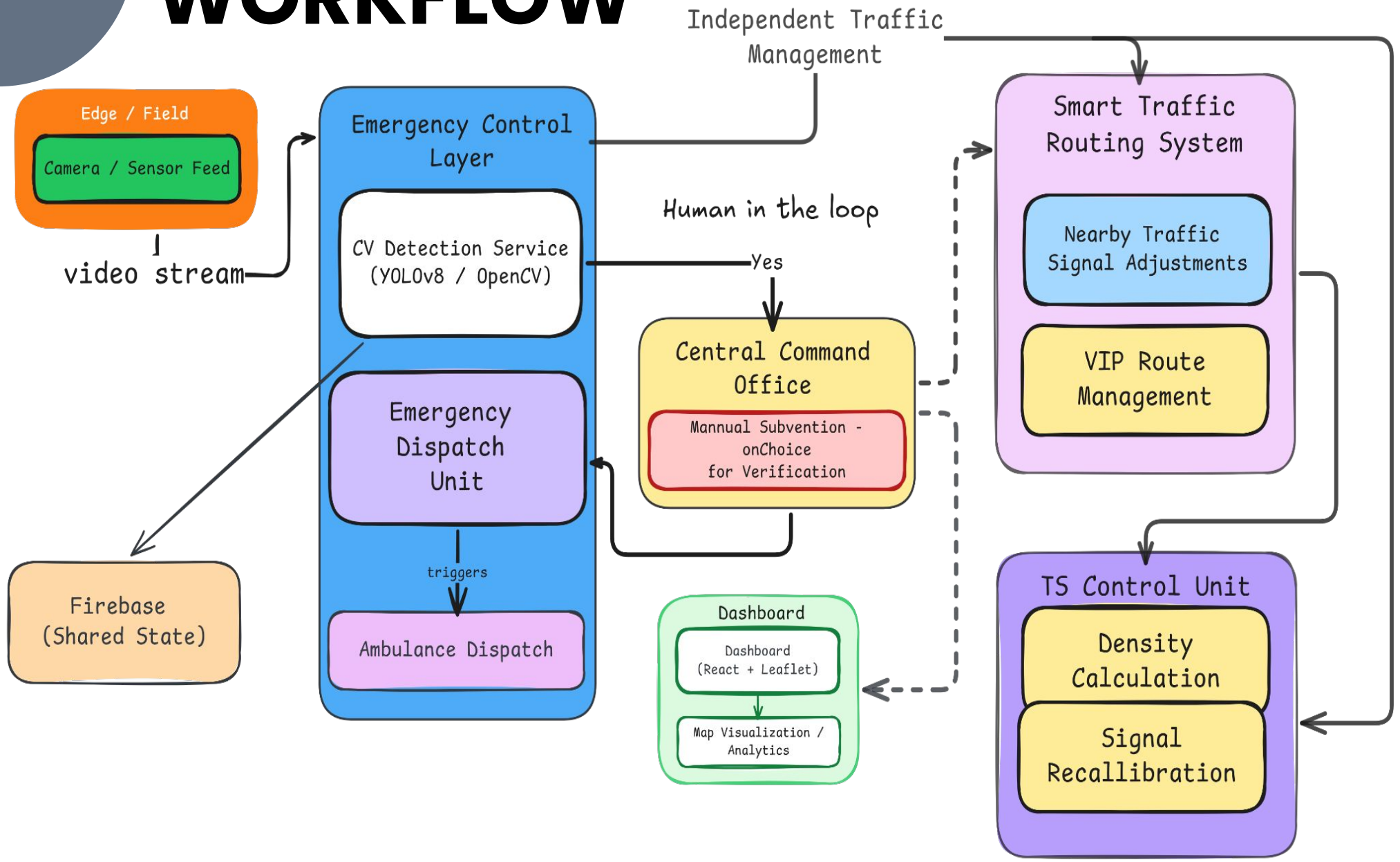




TRAFFIC SIMULATION



WORKFLOW


Independent Traffic Management
Edge / Field
Camera / Sensor Feed
video stream
Emergency Control Layer
CV Detection Service (YOLOv8 / OpenCV)
Human in the loop
Yes
Central Command Office
Mannual Subvention - onChoice for Verification
Emergency Dispatch Unit
triggers
Ambulance Dispatch
Firebase (Shared State)
Dashboard
Dashboard (React + Leaflet)
Map Visualization / Analytics
Smart Traffic Routing System
Nearby Traffic Signal Adjustments
VIP Route Management
TS Control Unit
Density Calculation
Signal Recallibration

Technologies Used

(Tech Stack Used)

Language: Typescript + React JS , Python

Libraries : **ultralytics**

Model: YoloV5 & YoloV8

Database: Firestore Firebase (NoSQL)

Frontend Stack: React JS + TailwindCss

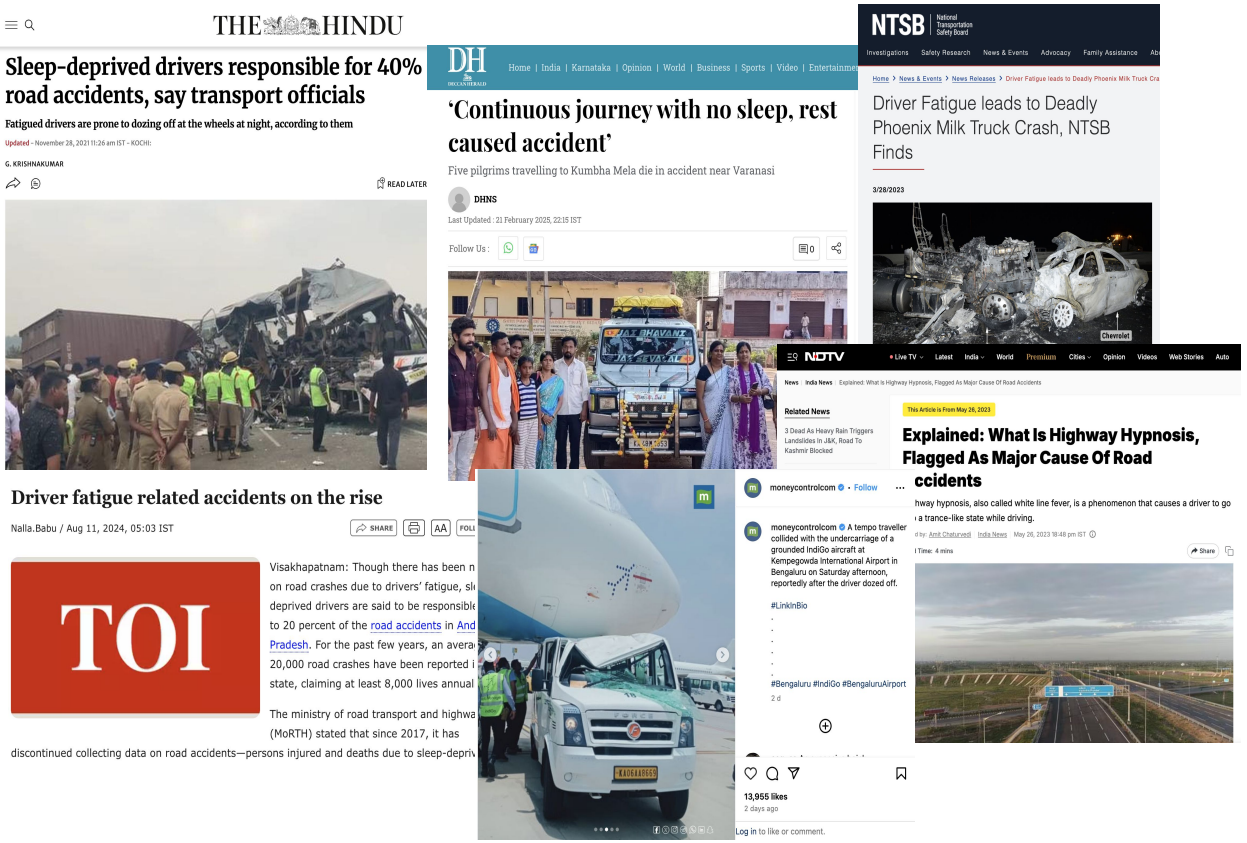
Road accident deaths rise to 1.77 lakh in 2024: Gadkari

ETV Bharat / Bharat

Traffic Mismanagement Bleeds India's Economy, Wastes Fuel, Chokes Cities: CSE Report

Complete Project Description Document ➡

https://docs.google.com/document/d/1ZMlPQJOoPbk-cGGREv_S0oQHsZYMOYUKn5lTC0iPKFs/edit?usp=sharing



Research & References

Road accident deaths rise to 1.77 lakh in 2024: Gadkari - The Hindu, accessed on April 6, 2026,	Morth data
India's Road Crisis 2025: Alarming Stats & Smart Reforms You Need... - CRECKK, accessed on April 6, 2026,	CRECKK
Traffic Detection Speed Comparison	getargus
Summary of ambulance response time data in the UK - Government Analysis Function, accessed on April 6, 2026,	ambulance response time data in the UK

Details of Team Members

Sr. No.	Name	Degree program	Year	Institution
1	HARSH RAJ SHUKLA	Btech Cse	3rd	Roorkee Institute of Technology
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4	ASHWANI RAJ	Btech Cse	3rd	Roorkee Institute of Technology